AI-Driven Predictive Analysis of Social Media Impact on Youth Mental Health with Chatbot Integration

A project report

Submitted in partial fulfilment of the requirements for the award of degree of

Bachelor of Computer Application

Submitted to

LOVELY PROFESSIONAL UNIVERSITY

PHAGWARA, PUNJAB

May 2025

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Declaration by student

To whom so ever it may concern

We Om Jaiswal, Anubhav Singh, Shiwaditya Chandra Mukesh, Harsh Rawat and Ashish Kumar Singh, hereby declare that the work done by us on "AI-Driven Predictive Analysis of Social Media Impact on Youth Mental Health with Chatbot Integration" under the supervision of Ms. Ranjit Kaur, Assistant Professor, School of Computer Application, Lovely Professional University, Phagwara, Punjab, is a record of original work for the partial fulfilment of the requirements for the award of the BCA, Bachelor of Computer Application.

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To whom so ever it may concern

This is to certify that Om Jaiswal, Anubhav Singh, Shiwaditya Chandra Mukesh, Harsh Rawat, Ashish Kumar Singh from Lovely Professional University, Phagwara, Punjab, has worked on "AI-Driven Predictive Analysis of Social Media Impact on Youth Mental Health with Chatbot Integration" under my supervision. It is further stated that the work carried out by the student is a record of original work to the best of my knowledge for the partial fulfilment of the requirements for the award of the BCA, Bachelor of Computer Application.

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TABLE OF CONTENT

ABSTRACT

As the world becomes smaller and smaller due to the occurrence of hand-held touchscreen cell phones with global networks right to use. Digital stalking and harassment have become common manifestations. When humans become anonymous, all pretence of responsibility is dropped, making human behaviour chaotic, leading to the effect of others' brain, emotional and spiritual conditions worsening, and making them choose disastrous options that include unaliving oneself to resolve the current predicament. Our study focuses on understanding the deleterious effect of digital media on young adults via data collection through Google survey forms input. Using fuzzy-based modelling to forecast deteriorating brain status like sleep disturbances. Sentimental analysis to determine the emotional status of a digital media user.

Chapter: - 1

INTRODUCTION

In the 21st century, social media was changed the way of we interact and communicate different from the early days where uses accessed the information by using the static websites. It's not just about scanning the information but it's about to being present in the digital era. Where digital tools help the people to share their opinions, thoughts, searching the information and interact with each other. Social media is a big platform where an individual connects with whole world, find solutions to every problem. These digital networks platform is not just about text-based interaction but carry a variety of tools like video, audio, live stream and as it forced on uses generated content and capability for uses to involve with each other in this collaborate and over-changing environment. There are some features such as like, comment, share and direct massaging, there features help in connecting with world and shaping in their networks. Emergence of online application and mobile apps like Facebook, WhatsApp, twitter and others has defined how business communication, personal relationship, marketing strategies, etc to be managed. These platforms provide a wide space where identities are built, people exchange their views and opinions, find what is right and wrong for them.

In the last, social media plays an important role in the life of billions of people, inspiring how we interact with each other. the digital shifts indicate a new beginning in the communication history. Social media is a tool which helps in transforming the behaviour of individual, a group of people or a society. In addition, the effect of social media increases beyond personal interaction – it plays a role a reframing education, business, politics and entertainment. Educational institution united platforms like YouTube and LinkedIn learning for knowledge sharing at the same tune business. Use Instagram for brand awareness and customer involvement. Politicians and actor use the power of platform such as Twitter(x) to support, spread consciousness and impact public discourse. The entertainment industry also depends on social media for marketing fan interaction and audience sharing. In each of these areas, the social media provide a fast spreading of ideas and a level of involvement that classical media cannot match. As digital technology continues to progress the purpose of social media is set to enhance, making it an essential part of 21st century.

As anything with having the influence of over billion of people from all over the world, different mind brings different ideas, different ideas bring certain conflict, conflict create tenson and tenson lead to drastic decisions. In past few years India had most death due to digital media, cause of death ranges from dangerous selfie accident, cyber harassment, cyber staking, cyber bulling leading to suicide, vitriolage, lynching due to foreign influence. This was still considered isolated and controllable with enough awareness and precaution. Then the question arises what influence is being plastered on the young mind. As visual contents are more interesting and increase the consumption by manyfold making mindless scrolling.

Chapter: - 2

LITERATURE REVIEW

The focus was mainly on establishing a link between the use of social media particularly when in excess, and some negative and positive psychological effects occurs. This describes the mounting evidence to suggest that the online engagement of different ideas can have harmful impacts on mental health mostly regarding anxiety, depression, FOMO and loneliness.

- Rawat and Amoli [1] (2023): Rawat and Amolis public health research presents wide evidence for the harmful psychological effects of social media addiction among various ages of work. The study highlight shows the use of social media is closely engaged with negative and positive emotions. This implies that it's not just how long one needs to spend on social media but also the character of that involvement once it becomes an addiction, then emotional distress is more likely to be the outcome. The author notably identifies the necessity of digital literacy and family engagement as being key strategies in showing these ill effects. Digital learning can help people learn how to use social media in a more effective and reflective manner while consequently helping them become aware of potential family engagement that highlights a role so that we can observe that families can play an important role in assisting and guiding their children for healthy social media behaviour. In case Rawat and Amoli highlight the need to solve social media's adverse psychological effects, which are showing through the education as well as having a supportive at-home environment.

- Rahman and Rahman [2] (2020): Focusing on university students at the University of Sharjah specifically, Rahman and his research show self-regulation of social media. The research finds that students who are unable to regulate their social media are likely to be sleep-deprived. That lack of sleep then leads to other issues, such as increasing or decreasing stress levels, and it also reflects on your academic performance. This study brings to the fore the real-world implications of unregulated societal issues on the lives of students. It is not just for a subjective feeling of unhappiness, but it is for the knowledge that you gain and implement for your better academic performance as well. The call for greater awareness is our personal responsibility: to be aware of the negative side of social media by Rahman and his colleague finds that individuals, with special address to young generations with problematic school atmospheres, need to be aware of the pros and cons of using social media and take immediate measures to manage their screen time. It involves such strategies as time limits, tracking apps and putting sleep routines ahead of late nights using mobile phones for chatting and browsing. Their findings emphasize academic achievement along with a healthy mental state.

Summary: The research studies are observed to be establishing a solid base for understanding the relationship between the two variables, namely, lack of self-control and excessive social media usage pointing towards the negative outwards in the psychological area. This research also talks about the addiction faced, coupled with poor self-regulation and other factors like the impact it has on emotional well-being and affected academic performance. Few remedies to counter, like online education, parental engagement and individual responsibility, lessen the importance of both society interventions to counter the possible negative effects of online platforms on mental health.

2.1 Sentimental and Behavior Analysis

This part mainly focuses on the conception of analysis, as in past simple correlations, and reasons for the various emotional meets of social networks, basically used to portion ways to sentiment analysis. It regards to user sentiment and their utilisation of, e.g., their utilisation of guarding their social network and give a fleeting exact viewpoint than simple positive or easy negative result. Sentiment analysis that enables researchers to get the emotional tone of online text data that is computationally, thus easing an understanding of how people behave and their feelings on social networks and other social network environments.

- Purohit and Mudgal [3] (2024) and Jadhav and Modhave [4] (2023): These are basically following an example of sentiment analysis to easily recognize user awareness of the main popular platforms like WhatsApp or Instagram. Through user-driven data (e.g. posts, comments, comments, comments, and comments, and responses to survey questions), both studies go beyond directly asking users for a simple yes or no if they like social media sooner and contribute to the inner emotional language users make use of to describe their engagement experience. Both studies report a similar outcome: although users acknowledge the benefits of communication on social media (connecting with others, sharing information, and staying connected), they often report associating social media with negative emotional states. Specifically, the authors identify stress as a common negative state associated with social media use, which could be linked to a few conditions, such as social comparison, fear of missing out (FOMO), and feeling pressure to uphold an online identity. Insecurity can materialize from worry about one’s body image and social validation (e.g., likes and comments).

Summary: This section illustrates the potential of sentiment analysis in revealing the emotional dimensions of social media experiences. It’s been suggested that users relate to platforms like WhatsApp and Instagram in nuanced and often ambivalent ways; while they acknowledge the advantages of communication, a considerable share of their experience is often associated with negative feelings like stress, insecurity and disrupted sleep. The distribution of sentiment toward neutral sentiment (or slightly negative) highlights the need to have a more analytic and refined understanding of the psychological consequences of social media, rather than relying upon the simple positive/negative storyline.

2.2 Technological Approaches and Innovations

The chapter is directed in a way towards the application of emergent for solving mental health problems using machine learning [ML] and natural language processing [NLP] for solving mental health problems that arise within social media spaces. Mainly, it explores possible methods for the earlier detection of mental illnesses and for the comprehension of them in an online setting.

- Kataru and King's [5] This research is as valuable for contributing towards technologies as it is to become action to monitor proactively mental health. They explored the uses of machine learning algos to find early indications of mental health issues. Their work is mainly focusing on accessing social indicators, which could be used with the child, who has been involved in different data points for recording their social connections and conduct. Social indicators could comprise attendance records from school participation in activities outside school, teacher opinion, and perhaps other gathered anonymised online activities. By training and working with the ML algorithm on their dataset, we find that social indicators established cases by psychological distress; they were able to achieve an accuracy of over 90% in identifying children at risk. Such high accuracy shows that ML has the potential to complement conventional mental health mobile screening procedures, especially in schools where early identification is critical. The large sample size of over 2600 students lends strength to their finding. This research demonstrates the accuracy and viability of AI-driven reports for mental health monitoring and paves the right way to start the bright future. Furthermore, there are also several concerns about data privacy, accountability in data processing and the correct way to use such technologies.

- Chandu et al. [6] (2024): Chandu and colleagues introduce a new concept with complementary data using (NLP) on open-source Reddit data. Being a public space with user-driven community discussions, Reddit facilitates a large database for content and user text that embodies lived experiences and feelings. NLP resources serve the purpose of enabling computers to learn and process human language and have proven valuable for researchers analyzing linguistic trends and emotional constructs on Reddit posts and comments Chandu et al. strongly highlighted the need for the identification of symptoms of anxiety and depression using language associated with them (e.g., sad and hopeless words, anxious words, changes in writing style). Their approach is particularly valuable as it taps into "the latest mental health problems that are not reported in clinical settings". Most individuals who have mental health issues might not present themselves for assistance due to various reasons such as stigma, accessibility, or just not knowing the possibility of accessing services. Evaluating online textual material, e.g., the kind of posts to be found in Reddit comments, can generate some understanding of such unreported cases and may serve as an early warning system while opening for intervention and support. In this article, the potential for utilising user-created content and NLP for achieving sense-making and reaching mental health at scale is highlighted with respect to populations who are online.

Summary: This chapter explain the immense potential that currently available for technological innovation has for treating mental health using concept of social media ML and NLP prove to be strong tools for the early identification of risk insights that are hidden inside the problem statement and assessments of risk. The work presented here shows how the technologies might identify at risk as we learn from online expressions of distress. But ethical issues regarding data privacy, correction rate of algorithms and use in line with good practice are of the highest priority and need careful attention as the technologies are involved and are used in the mental health setting.

2.3 Long term physical and emotional repercussions

In this part, researchers focus on the scope to consider the long-term and overall effect of too much use of digital media and screens extending beyond the short-term impact on the mind to include possible features that are benefits for physical health and long-term emotional burdens, especially in teenagers.

- Siahaan and Hanoum [7] (2023): The work of Siahaan and Hanoum provides critical evidence regarding the long-term effects of overexposure to screens. Longitudinal research, by following up on study participants for longer durations, allows the researchers to capture variations and make stronger associations compared to cross-sectional methods. Their evidence supports the association between excessive screen exposure and emotional disturbances, corroborating earlier work. They do go on, though, to show that these disturbances in emotion are not unique to mental health but may also "manifest in physical ailments". What this does is emphasise the interrelatedness of body and mind health and point out that excessive digital use can have concrete physical effects. The target physical ailments named hypertension, eye problems, and disturbed sleep patterns are serious health issues. Hypertension is itself a serious cardiovascular disease risk factor. Eye problems such as eye strain and dry eye are direct physical consequences of prolonged screen time. Disruptions to sleep cycles have long-term effects on physical and mental health, impacting mood, thinking, and internal body systems. Siahaan and Hanoum's findings staunchly "reinforce the value of digital balance and mindfulness." Digital balance is consciously restricting screen use and valuing activities offline as more important than screen time. Mindfulness suggests awareness of one's digital habits and their impact on those people who have well-being, which can mean a more planned and less compulsive engagement with the technology use. The longitudinal nature of their research generates robust evidence for the long-term consequences of overuse of screens and the importance of fostering healthy mental and digital habits towards emotional and physical health.

- Popat and Tarrant [8] (2023): Both the researchers' reviews mostly focus on lived experiences of social media and teenagers, and it gives a critical overview from different stakeholders. Qualitative study procedures allowed the researchers to tap into the intensity of what he/she faced and the complexity of the experience. By emphasizing young age viewpoints. Popat and Tarrant bring to fruition the tension and pressures of youth in present-day digital culture. Their abstract yield with adolescent social activity and development, peer validation, effectively illustrates adolescents effectively illustrates adolescents' acute awareness of the unattainable models of beauty represented on social media causing body image concern and self-esteem issues. Social media and FOMO as factors feeding chronic stress and physical anxiety. Adolescence is a stage of emotional and identical development, and so young people are especially vulnerable to the social comparisons, pressure and anxieties inherent in social media contexts.

Summary: In this research, researchers mainly focus on the long-term effects of high screen time on phones and social media platforms for both emotional and sentimental well-being. A longitudinal study that shows both sensitivity to bodily welfare and psychic health and takes cognizance of youths relatively vulnerable position towards longer-term damage through excessive use of digital activity, which is done on social media. It is used to reaffirm the need for imparting digital balance and awareness and the targeted people that form the basis of adolescents' health development in the digitalising world of the contemporary era.

Chapter: - 3

IMPLEMENTATION OF PROJECT

A dataset comprising 278 responses was collected through a Google Form that included questions about digital media usage and about their patterns, emotional states, sleep patterns and other relevant information. The analysis was done using two distinct methodologies:

- **Sentiment Analysis**: To understand emotional responses of people.
- **Fuzzy Logic-Based Prediction**: To estimate sleep-related issues as a stand-in for mental state.

3.1. Data assembly and Purification

The dataset was carefully assembled and cleaned or purified to make sure that high data quality and reliable analysis is achieved. The steps involved were, as follows:

a. NA Removal: Records of the data assembled with missing values in critical fields such as age, gender and sentiment-related answers were excluded. This made sure that the dataset was complete and suitable for smooth analysis of any kind.

b. Outlier Detection and Removal: Statistical limits were applied to identify outliers (e.g., absurd amounts of screen time or impossible age inputs) and remove or correct them.

c. Textual inputs were transformed into a standardized format using natural language preprocessing techniques such as lowercase transformation, punctuation removal and tokenization.

Time-related inputs like “late night,” “evening,” or “afternoon” were converted or transformed to 24-hour time values.

Numerical entries were scaled using standard score normalization when necessary.

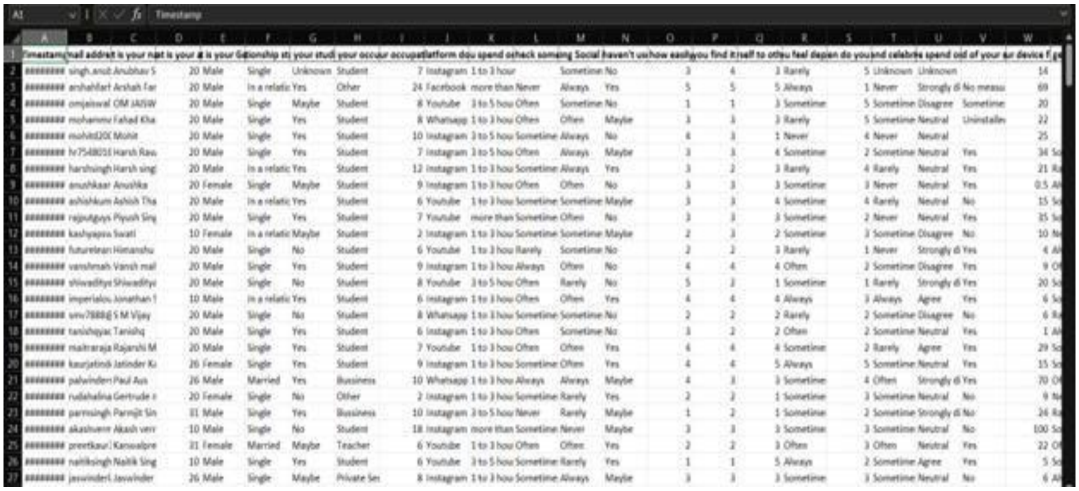


Figure 3.1.11: Image showing cleaned data in excel sheet

3.2. Analysis of Sentiment

a. Objective: The goal of this particular analysis in this study was to detect the emotional tone behind users' text responses, specifically how they feel about their digital media experiences. This qualitative insight forms a basis for understanding patterns in mental wellness.

b. Methodology:

Text Processing: The text responses were tokenized and vectorized using TF-IDF (abbr. as Term Frequency-Inverse Document Frequency) to give weightage to unique but significant terms.

Sentiment Classification: A lexicon-based approach using VADER (Valence Aware Dictionary and sentiment Reasoner) was employed to categorize responses into three classes: Positive, Neutral and Negative.

c. Visualization and Analysis: A sentiment distribution graph categorized by age groups was plotted to identify patterns across different age groups, which were: 0-10, 10-20, 20-26, 26-31 and 31-35.

A sentiment distribution graph categorized by gender was plotted to examine gender-based sentiment differences, namely male and female.

These visualizations helped identify which groups are most vulnerable to which impacts (positive, negative or both) of online digital media.

3.3. Fuzzy Rule System

a. Objective: To predict the likelihood of any responser facing sleep issues using a fuzzy inference system based on inputs such as screen time, time of social media usage, self-reported stress levels, and emotional states.

b. Fuzzy Logic Methodology:

Input Variables: screen time (in hours), comparison (how often does one compare themselves to digital medias people) and sleep issues.

Fuzzification: Inputs were converted into fuzzy sets using functions. For instance, screen time was categorized into Low (0-2 hrs), Medium (2-5 hrs) and High (5+ hrs).

Rule Base: A set of IF-THEN rules was designed. Example: IF screen time is High AND stress is High THEN sleep issue is High.

Defuzzification: The centroid method was used to convert the fuzzy output into a crisp value which indicates the level of sleep disturbance of a person.

3.4. Model Evaluation Metrics

To evaluate the accuracy of the predictions done for sleep issues using fuzzy logic, we used three metrics: Mean Squared Error (MSE), R-squared ® Score and Pearson Correlation Coefficient.

a. Mean Squared Error (MSE): MSE measures the average squared difference between actual and predicted values. Its mathematical formula to calculate the same is, as follows:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

MSE = mean squared error

n = number of data points

Y_i = observed values

$\hat{Y}_i$ = predicted values

Figure 3.4.1: Mathematical formula for MSE

b. R-squared Score: R evaluates how well the predictions approximate the actual data. Its mathematical formula to calculate the same is, as follows:

$$R^2 = 1 - \frac{RSS}{TSS}$$

R^2 = coefficient of determination

RSS = sum of squares of residuals

TSS = total sum of squares

Figure 3.4.2: Mathematical formula for R Squared Score

c. Pearson Correlation Coefficient: It quantifies the linear relationship between actual and predicted values. Its mathematical formula to calculate the same is, as follows:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

r = correlation coefficient

x_i = values of the x-variable in a sample

$\bar{x}$ = mean of the values of the x-variable

y_i = values of the y-variable in a sample

$\bar{y}$ = mean of the values of the y-variable

Figure 3.4.32:Mathematical formula for Pearson Correlation Coefficient

d. A value close to 1 implies strong positive correlation.

e. 3.5. Implementation Tools

f. Programming Language: Python

g. Libraries used

Pandas and NumPy for data manipulation

Matplotlib and Seaborn for visualization

Scikit-learn for evaluation metrics

NLTK and VADER for sentiment analysis

Skfuzzy for implementing fuzzy logic

This implementation successfully combined natural language processing with fuzzy logic to evaluate online digital media's influence on mental wellness. Sentiment analysis provided valuable qualitative insights, while fuzzy inference systems offered predictive data regarding sleep issues. The use of robust evaluation metrics ensured the reliability and interpretability of the predictive model. This approach provides a holistic framework to measure and understand the psychological effects of digital behavior.

3.6. Scope for future improvements

Integration of deep learning models to improve sentiment classification accuracy.

Expansion of fuzzy logic rules to cover more mental health indicators such as anxiety and deteriorating attention span.

Deployment of a real-time application to track and visualize ongoing mental health trends based on user inputs.

Code:

For data cleaning

```
import pandas as pd

# Load the dataset

df = pd.read_csv("Social Media affecting Mental Health (Responses) - Form responses
1.csv")


# Step 1: Drop unnecessary columns and rename others for simplicity

df = df.drop(columns= ['Timestamp', 'Email address', 'What is your name?'])

df.columns = [

    'Age', 'Gender', 'Relationship_Status', 'Jeopardized_Life',

    'Occupation', 'Work_Hours', 'Common_Platform', 'Daily_SM_Hours',

    'Unintended_Usage', 'Purposeless_Usage', 'Restlessness',

    'Distraction_Level', 'Concentration_Difficulty', 'Comparison_Level',

    'Depression_Frequency', 'Sleep_Issue_Level', 'Appearance_Effect',
```

```
    'Hiding_Usage', 'Tried_To_Quit', 'Weekly_SM_Hours', 'Quit_Frequency',
    'Quit_Measures', 'Other_Symptoms'
]
# Step 2: NA Removal
primary_fields = ['Daily_SM_Hours', 'Comparison_Level', 'Sleep_Issue_Level']
df = df.dropna(subset=primary_fields)

# Step 3: Convert to numeric and drop conversion errors
df['Daily_SM_Hours'] = pd.to_numeric(df['Daily_SM_Hours'], errors='coerce')
df['Comparison_Level'] = pd.to_numeric(df['Comparison_Level'], errors='coerce')
df['Sleep_Issue_Level'] = pd.to_numeric(df['Sleep_Issue_Level'], errors='coerce')
df = df.dropna(subset=primary_fields)

# Step 4: Outlier Detection and Removal
df = df[
    (df['Daily_SM_Hours'] >= 0) & (df['Daily_SM_Hours'] <= 24) &
    (df['Comparison_Level'] >= 1) & (df['Comparison_Level'] <= 5) &
    (df['Sleep_Issue_Level'] >= 1) & (df['Sleep_Issue_Level'] <= 5)
]
# Step 5: Standardization (already in hours and scale of 1–5)
df['Daily_SM_Hours'] = df['Daily_SM_Hours'].astype(float)
df['Comparison_Level'] = df['Comparison_Level'].astype(float)
df['Sleep_Issue_Level'] = df['Sleep_Issue_Level'].astype(float)

# df is now clean and ready for fuzzy logic modeling
print(df.head())
```

For Fuzzy Logic Prediction

Code

```
import pandas as pd

import numpy as np

import skfuzzy as fuzz

from skfuzzy import control as ctrl

from sklearn.metrics import mean_squared_error, r2_score


# Step 1: Load data

df = pd.read_excel("Social_Media_Cleaned.xlsx")


# Step 2: Clean and map hours

hours_mapping = {

    'Less than an hour': 0.5,

    '1 to 3 hour': 2,

    '3 to 5 hour': 4,

    'more than 5 hour': 6

}

columns = {

    'hours_per_day': 'How many hours do you spend on social media in a day?',

    'comparison': '15. On a scale of 1-5, how often do you compare yourself to other
successful people through the use of social media?',

    'sleep_issues': 'On a scale of 1 to 5, how often do you face issues regarding sleep?'

}

df_clean = df[list(columns.values())].rename(columns= {v: k for k, v in
columns.items()})

df_clean['hours_per_day'] = df_clean['hours_per_day'].map(hours_mapping)

df_clean['comparison'] = pd.to_numeric(df_clean['comparison'], errors='coerce')
```

```
df_clean['sleep_issues'] = pd.to_numeric(df_clean['sleep_issues'], errors='coerce')
df_clean.dropna(inplace=True)

# Step 3: Define fuzzy logic variables
hours = ctrl.Antecedent(np.arange(0, 7, 0.1), 'hours')
comparison = ctrl.Antecedent(np.arange(1, 6, 1), 'comparison')
sleep_issues = ctrl.Consequent(np.arange(1, 6, 1), 'sleep_issues')

# Membership functions
hours['low'] = fuzz.trimf(hours.universe, [0, 0, 2])
hours['medium'] = fuzz.trimf(hours.universe, [1, 3, 5])
hours['high'] = fuzz.trimf(hours.universe, [3, 6, 6])

comparison['rare'] = fuzz.trimf(comparison.universe, [1, 1, 2])
comparison['sometimes'] = fuzz.trimf(comparison.universe, [2, 3, 4])
comparison['often'] = fuzz.trimf(comparison.universe, [3, 5, 5])

sleep_issues['none'] = fuzz.trimf(sleep_issues.universe, [1, 1, 2])
sleep_issues['moderate'] = fuzz.trimf(sleep_issues.universe, [2, 3, 4])
sleep_issues['severe'] = fuzz.trimf(sleep_issues.universe, [4, 5, 5])

# Step 4: Define rules
rule1 = ctrl.Rule(hours['low'] & comparison['rare'], sleep_issues['none'])
rule2 = ctrl.Rule(hours['medium'] & comparison['sometimes'], sleep_issues['moderate'])
rule3 = ctrl.Rule(hours['high'] | comparison['often'], sleep_issues['severe'])

sleep_ctrl = ctrl.ControlSystem([rule1, rule2, rule3])
```

```
sleep_sim = ctrl.ControlSystemSimulation(sleep_ctrl)

# Step 5: Predict using fuzzy logic
predicted_sleep = []
for _, row in df_clean.iterrows():
    try:
        sleep_sim.input['hours'] = row['hours_per_day']
        sleep_sim.input['comparison'] = row['comparison']
        sleep_sim.compute()
        predicted_sleep.append(sleep_sim.output['sleep_issues'])
    except:
        predicted_sleep.append(np.nan)

df_clean['predicted_sleep_issues'] = predicted_sleep

# Step 6: Print sample table
print("\n📊 Sample Prediction Table:")
print(df_clean[['hours_per_day', 'comparison', 'sleep_issues',
'predicted_sleep_issues']].head(10).to_string(index=False))

# Step 7: Accuracy analysis (handling NaNs)
valid_df = df_clean.dropna(subset= ['sleep_issues', 'predicted_sleep_issues'])

actual = valid_df['sleep_issues']
predicted = valid_df['predicted_sleep_issues']

mse = mean_squared_error(actual, predicted)
r2 = r2_score(actual, predicted)
```

```
corr = actual.corr(predicted)

print(f"\n📈 Mean Squared Error (MSE): {mse:.2f}")
print(f"🔁 R² Score: {r2:.2f}")
print(f"📌 Correlation between Actual and Predicted: {corr:.2f}")

# Step 8: Save to Excel
df_clean.to_excel("Predicted_Sleep_Issues_Output.xlsx", index=False)
print("\n✅ Final output saved as 'Predicted_Sleep_Issues_Output.xlsx'")
```

For Sentimental Analysis

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the data
file_path = "Social_Media_Cleaned.xlsx"
df = pd.read_excel(Social_Media_Cleaned.xlsx)

# Convert 'Age' to numeric, ignore errors
df['Age'] = pd.to_numeric(df['Age'], errors='coerce')
df = df.dropna(subset=['Age']) # Remove rows with invalid/missing age

# Set plot style
sns.set(style="whitegrid")
```

```
import matplotlib.pyplot as plt

# Visualization 1: Average Sentiment by Age
plt.figure(figsize= (12, 6))
plt.bar(sentiment_by_group['What is your age?'], sentiment_by_group['sentiment_score'],
color='skyblue')
plt.xlabel('Age')
plt.ylabel('Average Sentiment Score')
plt.title('Average Sentiment Score by Age')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.savefig('sentiment_by_age.png')
plt.show()

# Visualization 2: Average Sentiment by Gender
plt.figure(figsize= (8, 6))
sentiment_by_gender = sentiment_by_group.groupby('What is your
Gender')['sentiment_score'].mean().reset_index()
plt.bar(sentiment_by_gender['What is your Gender'],
sentiment_by_gender['sentiment_score'], color=['lightcoral', 'lightgreen', 'lightskyblue'])
plt.xlabel('Gender')
plt.ylabel('Average Sentiment Score')
plt.title('Average Sentiment Score by Gender')
plt.tight_layout()
plt.savefig('sentiment_by_gender.png')
plt.show()
```

Chapter 4:

RESULT AND DISCUSSION

All experiments were run in a Python-based setup on a 64-bit Windows 10 OS, driven by an Intel(R) Core (TM) i5 processor and 8 GB of RAM. Preprocessing of the textual responses was succeeded by sentiment labeling, and fuzzy logic was used to examine the effect of usage behavior on sleep disturbances.

The central features employed in the study were

- Gender and Age
- Daily hours on social media Comparison tendencies
- Reported levels of sleep disturbance Overall sentiment scores

From the sentiment distribution analysis (Figure 4), Neutral sentiments were most dominant, especially among male respondents (59 males vs. 40 females). Yet, female users exhibited slightly more positive sentiment responses than males.

Age-wise sentiment analysis (Figure 4.2) showed more variation in overall sentiment scores among older subjects (ages 31 and 35), while younger groups (ages 10–26) showed more homogeneous levels of sentiment, with the majority being in the neutral area and a couple of outliers for both positive and negative polarity.

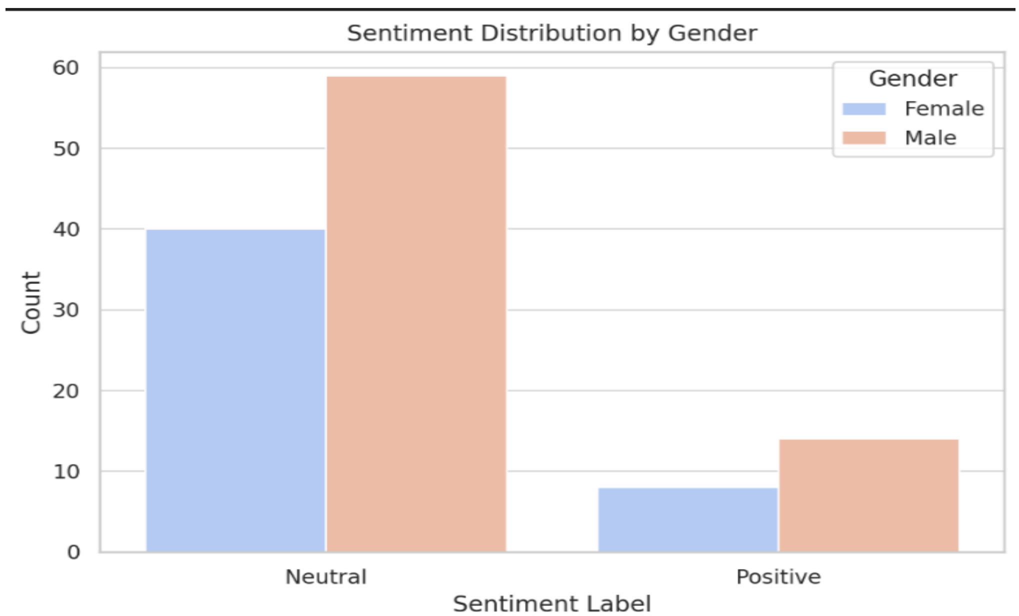
```
📊 Sample Prediction Table:
 hours_per_day  comparison  sleep_issues  predicted_sleep_issues
           2.0           3             5                3.000000
           6.0           5             1                4.666667
           4.0           3             5                3.426426
           2.0           3             5                3.000000
           4.0           1             4                4.577778
           4.0           4             2                4.611111
           2.0           3             4                3.000000
           2.0           3             3                3.000000
           2.0           4             4                4.611111
           6.0           3             2                4.666667

📈 Mean Squared Error (MSE): 3.50
🔁 R² Score: -1.09
📌 Correlation between Actual and Predicted: 0.14

✅ Final output saved as 'Predicted_Sleep_Issues_Output.xlsx'
```

4. 1 Prediction based on sleeping issues by the social media users

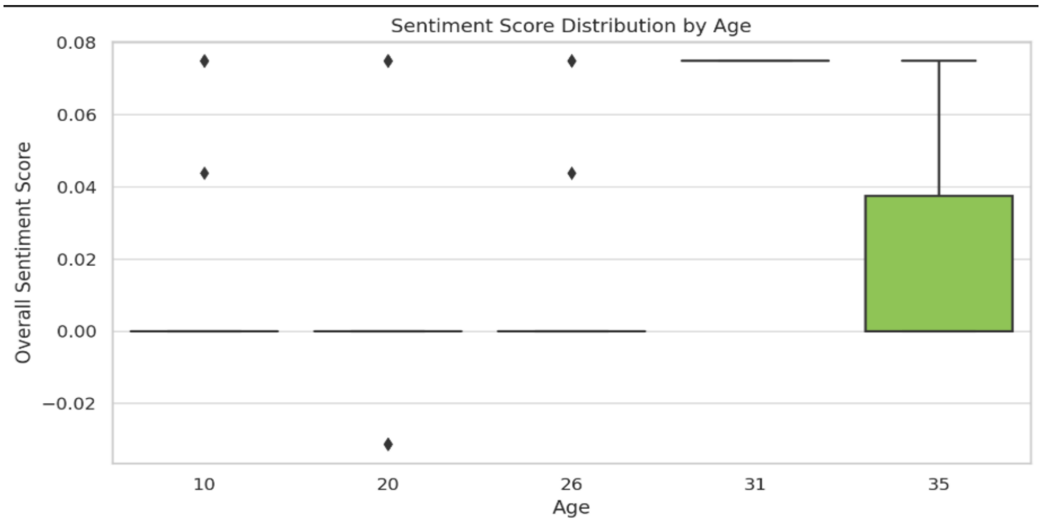
The fuzzy system was employed to predict sleep issues from numerical behavioral trends such as screen time and self-comparison levels. The prediction outputs of the model are shown in Figure 4.3. Some of the outputs are explained as follow:

Mean Squared Error (MSE): 3.50, which represents a moderate level of difference between actual data and predicted data for sleep issues. R^2 Score: -1.09, which means that it is a bad fit of the model that does not account for variance. Correlation (Predicted VS Actual): 0.14, which indicates a weak linear relationship, meaning that the model does not generalise well across the sample.

These results indicate that while fuzzy modelling generates an understandable model to identify high-risk users, it overestimates sleep issues, especially among heavy users or highly social comparative users. Despite the limitations of accuracy, the rule-based method provides a valuable first-line diagnostic tool for mental illness screening via social media analysis.



4. 2 Sentiment distribution by gender



4. 3 Sentiment distribution by Age

Website:

Name: Connect

Aim: To give users better feedback on their assessment and a platform for easier understanding of our result regarding "AI-Driven Predictive Analysis of Social Media Impact on Youth Mental Health with Chatbot Integration"

We have prepared set of web page for user to interact with our study, asses their mental state based of our study, have AI enabled chatbot for better understand and discuss their result, also to get better suggestions to improve their digital media habits and resources to get going for better habits.

Websites consist of:

- Home Page
- Research page
- Resource hub page
- Chat page
- Assessment page

Home Page:
It contains a brief introduction about the website and all the navigation in a Docker format for better user experience. From this page you can navigate to all available pages.

Research page:
It contains the result and discussion for the study and graphs and charts for better and clear understanding of our study. It also contains our survey data for others' aid.

Chat page:
Page available with a chat interface for chatting with the Gemini (AI large language model)-powered chatbot for user conversation. Conversation is not saved for user privacy.

Resource Hub page:
This page is about a few available resources available for users to start their digital betterment journey. This includes the Digital Wellbeing app, a few articles, books and organisations.

Assessment Page:
A quiz-like page for users to answer and get the assessment result of how good engagement there is with their digital media and their mental health. Generic 23 questions are there for the user to answer; from there, it concludes the assessment result criteria of engagement, emotion, comparison, cognition, relationships, coping and values. After that, the user is provided with recommended next steps for users to take to improve their mental state.

Home Page:

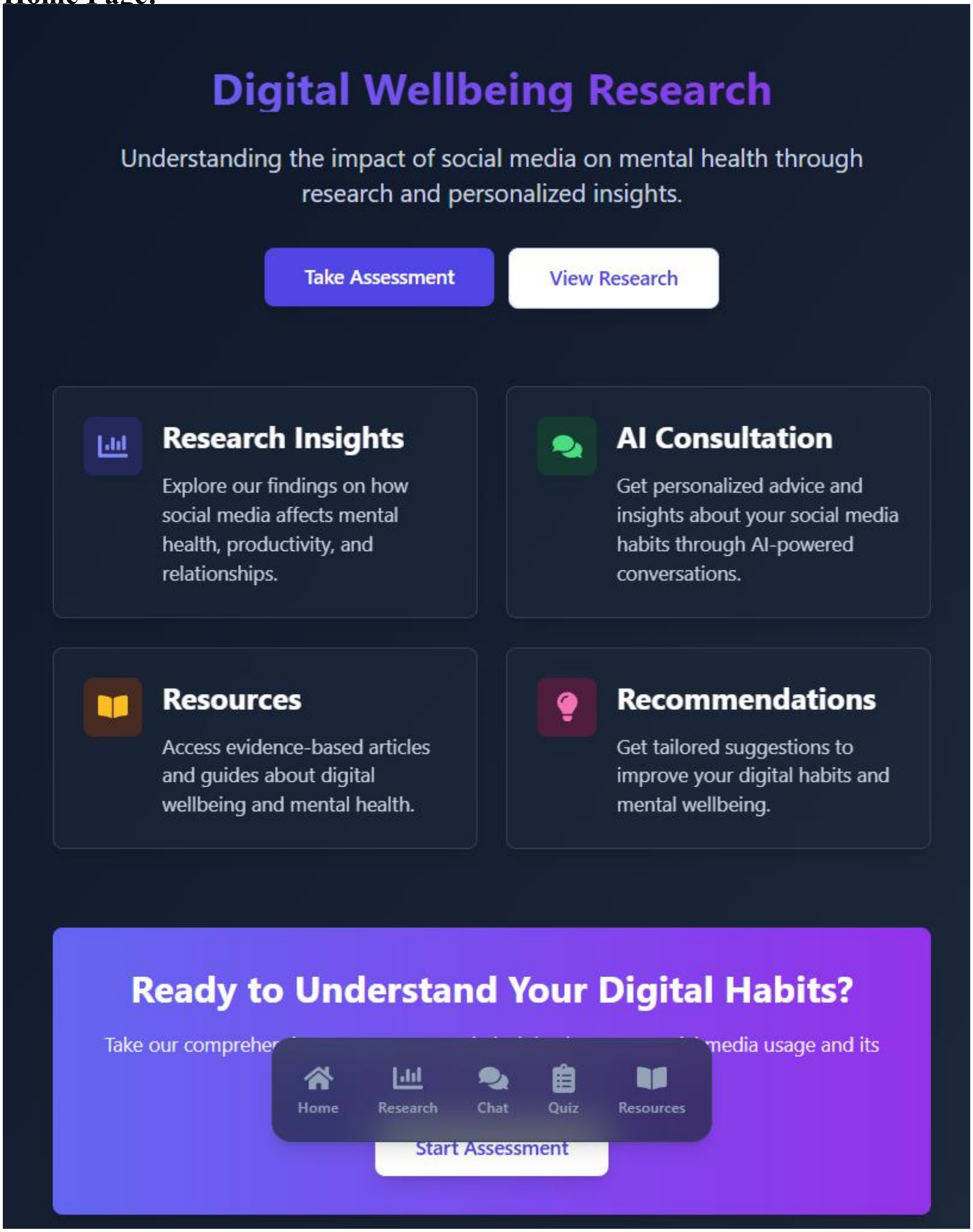


Figure 4.4 Home page

Research Page:

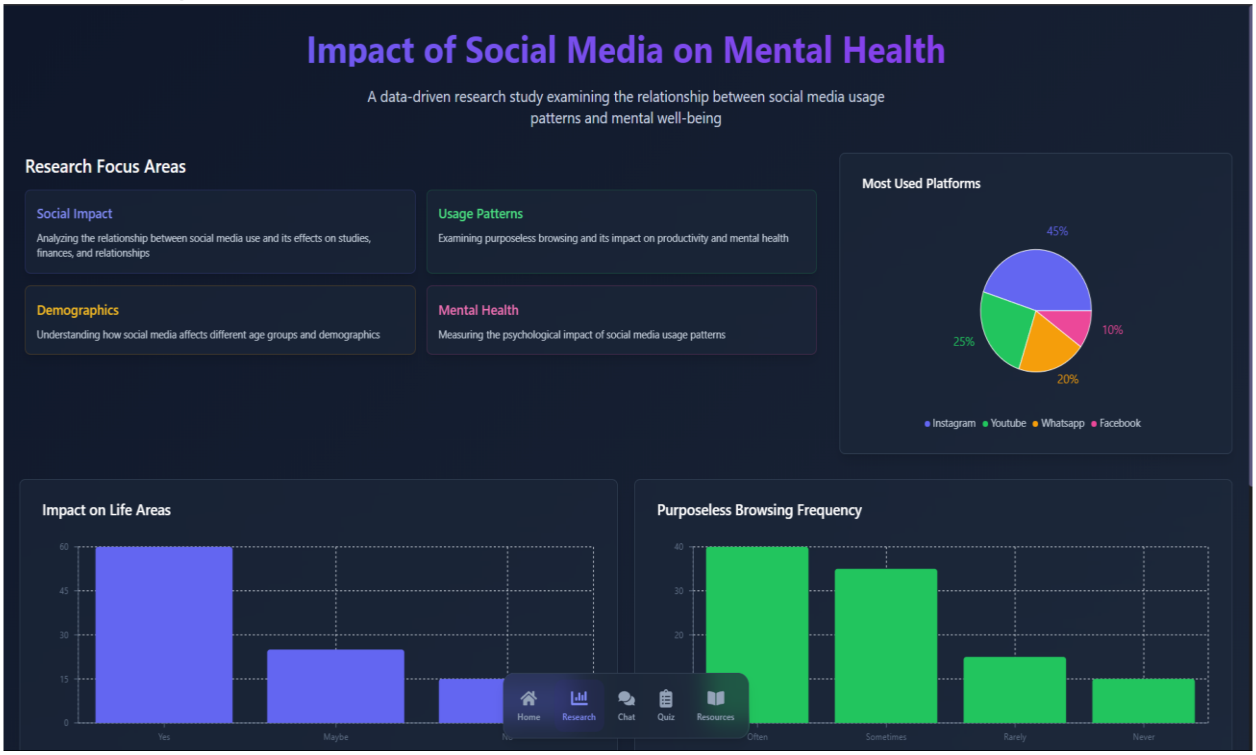


Figure 4.5 Research page

Chat Page:

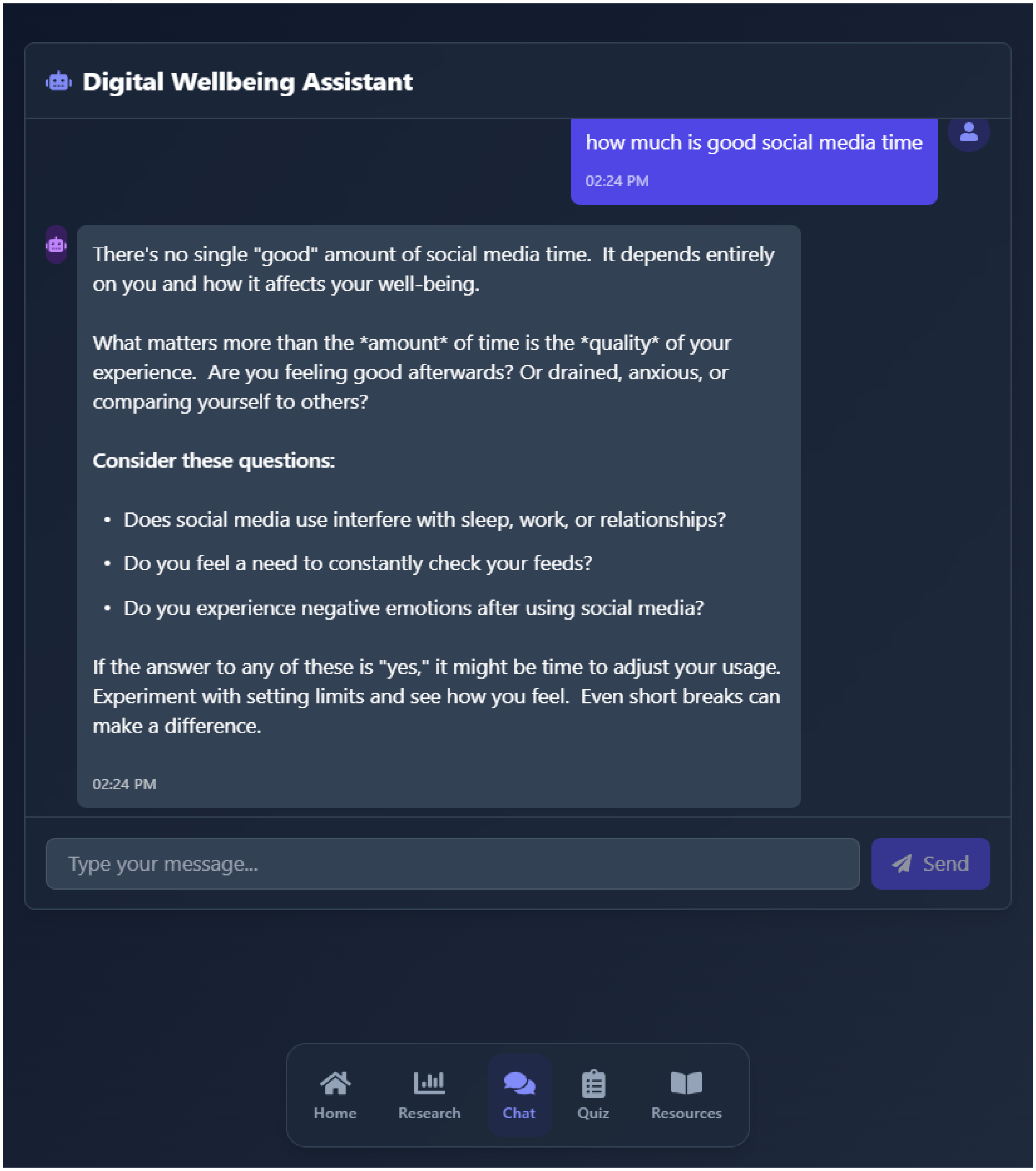


Figure 4.6 Chat page

Assessment Page:

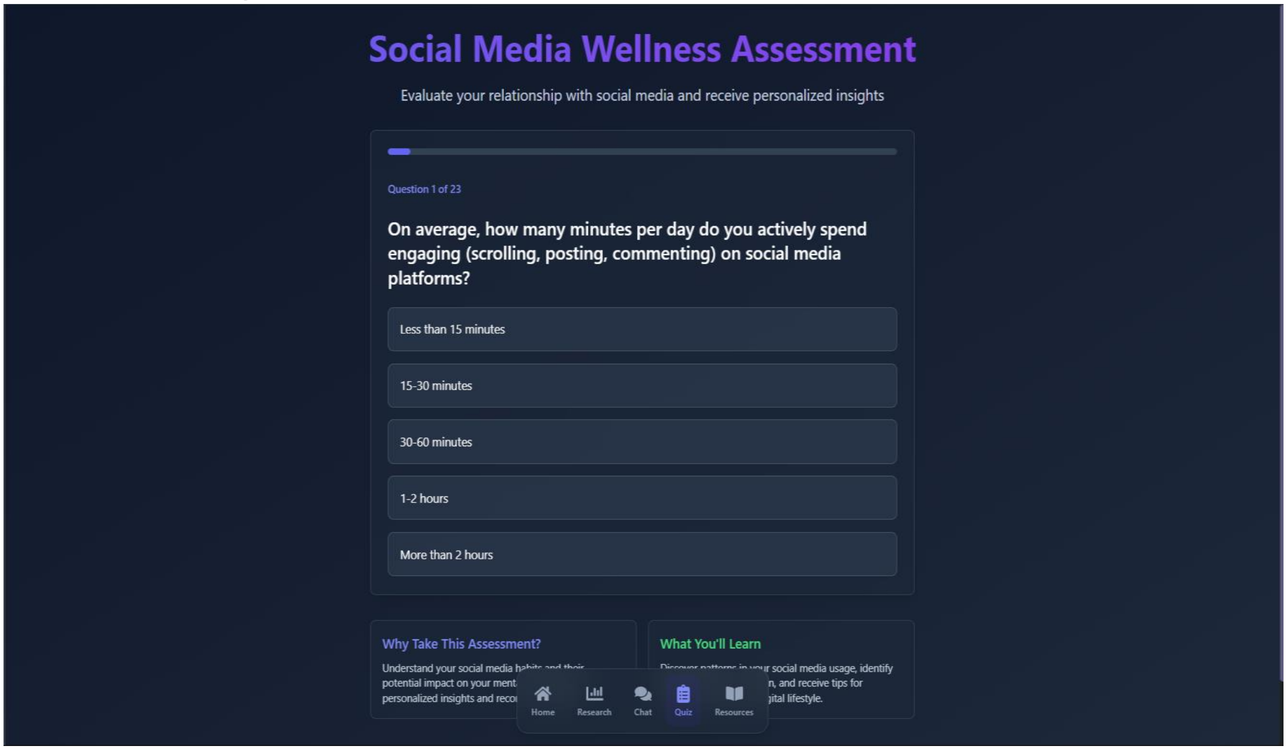


Figure 4.7 Assessment page (1)

Figure 4.7 Assessment page (2)

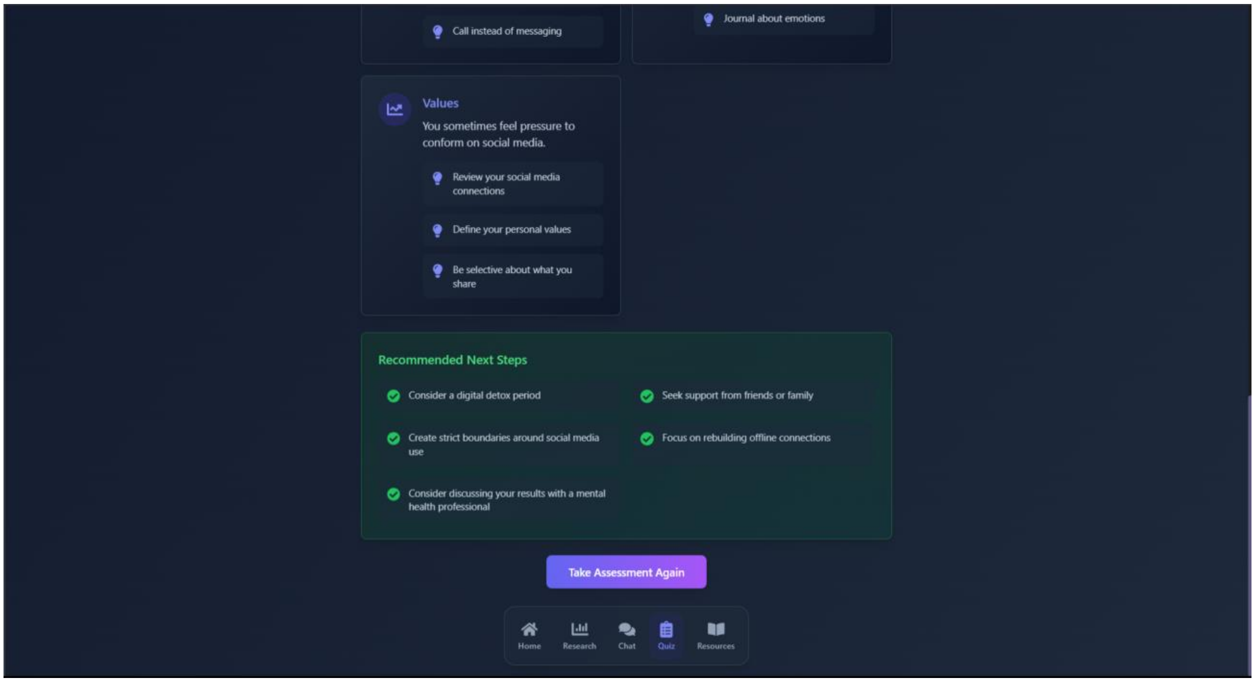


Figure 4.7 Assessment page (3)

Figure 4.7 Assessment page

Resource Hub Page:

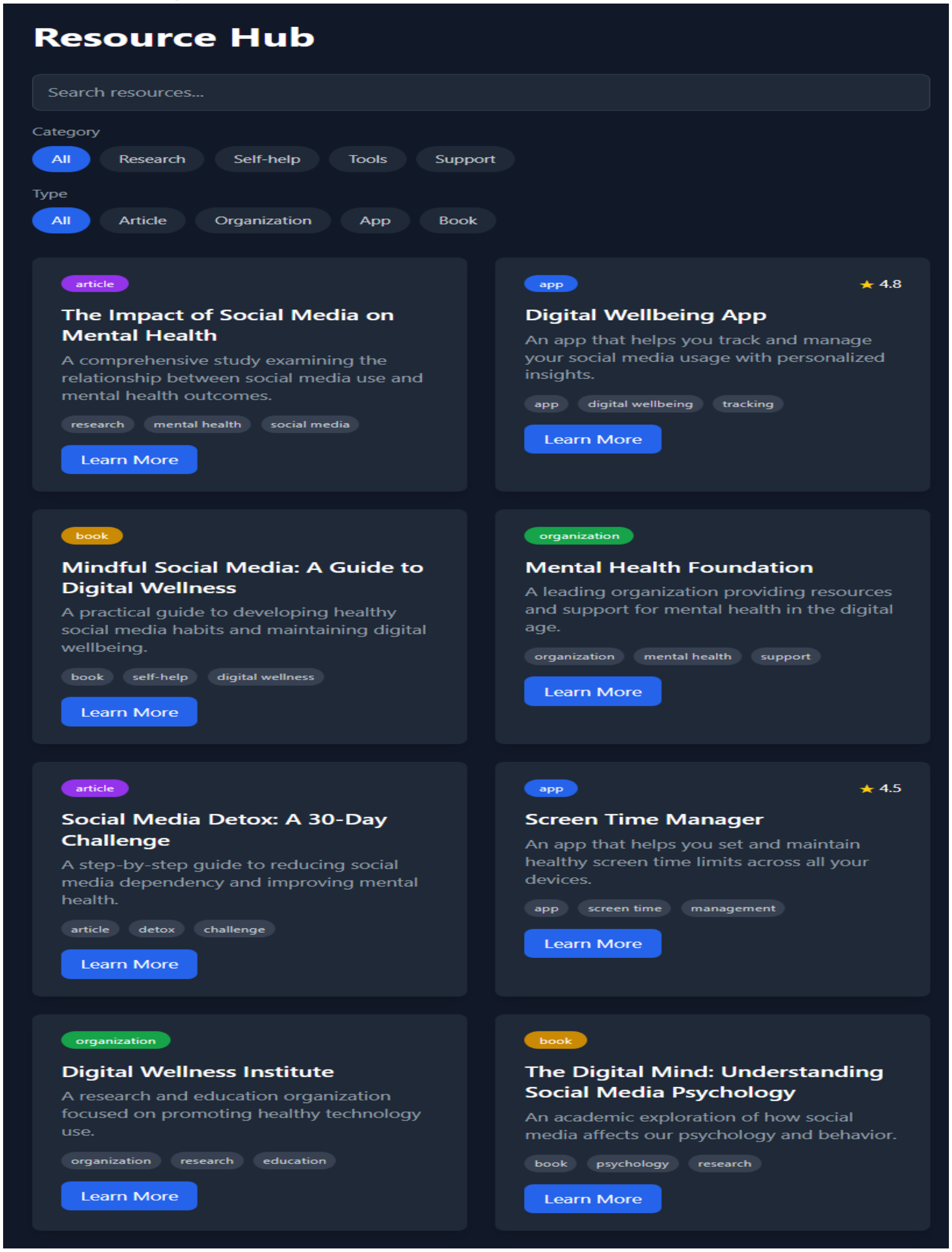


Figure 4.8 Resource Hub Page

Chapter: - 5

CONCLUSION

This research investigated the psychological effects of social media by analysing sentiment and predicting sleep disturbance based on fuzzy logic from self-reported information gathered using Google Forms. The dataset captured significant behavioural characteristics, emotional dispositions, and trends of digital media usage. The goal was to determine if subjective patterns of sentiment and quantitative behavioural metrics could be a reliable indicator of potential mental health issues, especially sleep problems.

Sentiment analysis indicated that many of the respondents were in neutral emotional states, with female users showing a weak positive sentiment inclination compared to males. Sentiment analyses that were distributed by age showed greater emotional variation among older respondents, with younger segments registering relatively stable and neutral sentiment levels.

A fuzzy inference system was then applied to predict sleep difficulties using predictors such as daily screen exposure and levels of self-comparison. Although the model achieved a Mean Squared Error (MSE) of 3.50 along with the R^2 value of -1.09 and a low correlation of 0.14 between predicted and actual values, suggesting that the model lacks good generalisability across the whole dataset. The model tended to overpredict risk, particularly among individuals with high social media use and high comparison tendencies.

Despite these limitations, the fuzzy logic model provided a comprehensible and interpretable structure for the analysis of behavioural health data. It offers a rule-based foundation for the evaluation of mental well-being and can be implemented in mobile apps or feedback APIs. Such systems hold promise for early intervention and tailored communication between mental health clinicians and users, in the service of the larger goal of real-time digital tracking of mental health.

PUBLICATION DETAILS

Gmail **Ranjit Kaur Walia <ranjitwalia2025@gmail.com>**

Submission of Book Chapter for Book - Fuzzy Logic for Healthcare and Biomedical Applications
1 message

Ranjit Kaur Walia <ranjitwalia2025@gmail.com> Thu, May 1, 2025 at 11:48 PM
To: "editedbooks.chapters@gmail.com" <editedbooks.chapters@gmail.com>

Dear Sir/Ma'am

I hope this email finds you well.

Please find attached the abstract of my proposed chapter titled **"Integrating Fuzzy Rule based Inference System and Sentiment Analysis to Measure Social Media's Influence on Mental Well-Being of People"** for the upcoming book titled ***"Fuzzy Logic for Healthcare and Biomedical Applications."*** I am excited about the opportunity to contribute to this significant publication and look forward to your kind consideration.

Please let me know if you require any further clarification or changes.

1Updated1_Book_Chapter_Fuzzy.pdf
647K

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Topic Approval

TOPIC APPROVAL PERFORMA

School of Computer Application (SCA)

Program : P124::BCA

COURSE CODE : CAP463 **REGULAR/BACKLOG :** Regular **GROUP NUMBER :** CARGC0008

Supervisor Name : Ranjit Kaur **UID :** 28632 **Designation :** Assistant Professor

Qualification : ______________________ **Research Experience :** ______________________

SR.NO.	NAME OF STUDENT	Prov. Regd. No.	BATCH	SECTION	CONTACT NUMBER
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2	Anubhav Singh	12208856	2022	D2203	7380718141
3	Shiwaditya Chandra Mukesh	12208840	2022	D2203	7255902049
4	Om Jaiswal	12207638	2022	D2203	7001581538
5	Harsh Rawat	12207447	2022	D2203	9643403819

SPECIALIZATION AREA : Databases **Supervisor Signature:** ___________________

PROPOSED TOPIC : AI-Driven Predictive Analysis of Social Media's Impact on Youth Mental Health with Chatbot Integration

Qualitative Assessment of Proposed Topic by PAC		
Sr.No.	**Parameter**	**Rating (out of 10)**
1	Project Novelty: Potential of the project to create new knowledge	7.51
2	Project Feasibility: Project can be timely carried out in-house with low-cost and available resources in the University by the students.	7.54
3	Project Academic Inputs: Project topic is relevant and makes extensive use of academic inputs in UG program and serves as a culminating effort for core study area of the degree program.	8.00
4	Project Supervision: Project supervisor's is technically competent to guide students, resolve any issues, and impart necessary skills.	8.02
5	Social Applicability: Project work intends to solve a practical problem.	7.54
6	Future Scope: Project has potential to become basis of future research work, publication or patent.	7.54

PAC Committee Members		
PAC Member (HOD/Chairperson) Name: Dr. Sartaj Singh	UID: 11303	Recommended (Y/N): Yes
PAC Member (Allied) Name: Dr. Pawan Kumar	UID: 11522	Recommended (Y/N): Yes
PAC Member 3 Name: Jasleen Kaur Paintal	UID: 30857	Recommended (Y/N): Yes

Final Topic Approved by PAC: **AI-Driven Predictive Analysis of Social Media's Impact on Youth Mental Health with Chatbot Integration**

Overall Remarks: Approved

PAC CHAIRPERSON Name: 32898::Dr. Anand Kumar Shukla **Approval Date:** 03 Apr 2025

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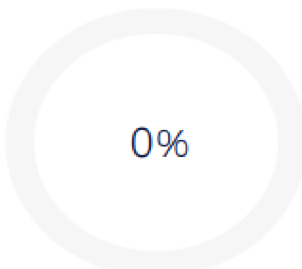

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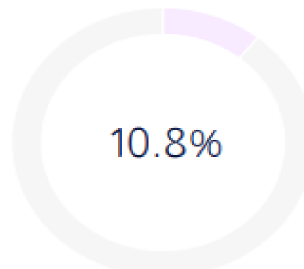

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